

AlphaCom Telecommunications

Customer Churn Prediction

CAPSTONE PROJECT REPORT

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PREDICTION

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1. Understanding the Business Problem

1.1 Problem Statement

AlphaCom, a leading telecommunications provider, is experiencing a significant increase in customer churn despite offering competitive services and a diverse product portfolio. The lack of insight into the underlying causes makes it difficult to identify at-risk customers early, resulting in reactive rather than proactive retention efforts.

1.2 Need Of The Study

Acquiring a new customer costs five to seven times more than retaining an existing one. Without a data-driven approach, AlphaCom cannot efficiently allocate retention resources or intervene before customers leave. This study addresses that gap by building a predictive model to identify high-risk customers early.

1.3 Business Opportunity

By using predictive analytics and machine learning, telecom companies can improve customer retention, reduce revenue loss, and make data-driven business decisions to enhance customer satisfaction and long-term profitability.

1.4 Social Opportunity

Ensure continued access to essential communication services for vulnerable groups such as senior citizens and rural customers who depend heavily on reliable connectivity.

2. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the dataset analyse patterns and relationships between variables, and generate meaningful business insights before model building.

2.1 Data Collection And Background

The dataset contains telecom customer information related to demographics, service subscriptions, and customer churn status. It includes both categorical and numerical variables. The target variable, Churn, indicates whether a customer has discontinued the company's services.

2.2 Data Overview

Attribute	Details
Total Records	12,055
Total Features	20 (19 predictors + 1 target)
Target Variable	Churn (Yes/No)
Churn Rate	28.3% (Yes) 71.7% (No)
Numeric Features	tenure, MonthlyCharges, TotalCharges
Categorical Features	17 (demographics, services, billing)

Table 2.2 – Data Overview

2.3 Feature Categories

The dataset contains 21 features (20 predictors and 1 target variable) categorized into demographic, account, service, billing, and target-related attributes. A detailed feature-wise data dictionary is provided in Appendix A.

Feature Category	Variables
Demographics	Gender, SeniorCitizen, Partner, Dependents
Account Details	tenure, Contract, PaymentMethod, PaperlessBilling
Phone Services	PhoneService, MultipleLines
Internet & Support Services	InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies
Billing Information	MonthlyCharges, TotalCharges
Target	Churn

Table 2.3 – Feature Categories

‘TotalCharges’ and ‘MonthlyCharges’ were initially imported as object variables due to the presence of currency symbols and invalid entries.

2.4 Statistical summary

Variable	count	mean	std	min	25%	50%	75%	max
SeniorCitizen	12055	0.12	0.32	0	0	0	0	1
tenure	11451	31.24	25.03	-3	6	28	54	74
MonthlyCharges	11754	64.37	30.33	15.29	30.31	71.35	89.38	121.67
TotalCharges	10850	2291.76	2277.46	-197	383.65	1328.75	3959.08	9039.92

Table 2.4 – Statistical Summary

During initial data exploration, negative values were identified in tenure and TotalCharges. These were treated as anomalous entries and converted to missing values (NaN). To maintain the integrity of the evaluation and prevent data leakage, imputation was performed only after the train-validation-test split, ensuring that information from the test set did not influence the training process.

2.5 Univariate analysis

2.5.1 Distribution of Numerical Variables

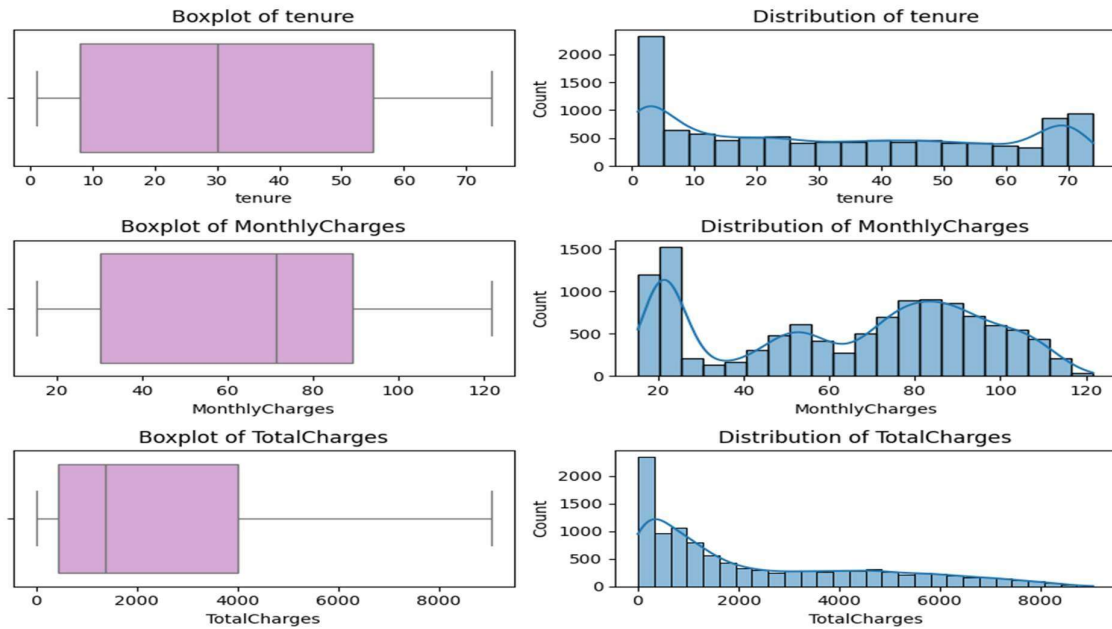


Fig 2.5.1 – Distribution of Numerical Variables

Variable	Distribution	Observation	Business Insight
tenure	Bimodal / Ushaped	Peaks at 0–10 and 65–72 months	New customers show higher churn risk
MonthlyCharges	Multimodal	Clear low, mid, and high pricing tiers	Customers fall into distinct pricing segments
TotalCharges	Right-Skewed	Majority concentrated at lower values	Few long-term customers contribute high revenue

Table 2.5.1 – Univariate Analysis – Numerical Variables

2.5.2 Distribution of Categorical Variables

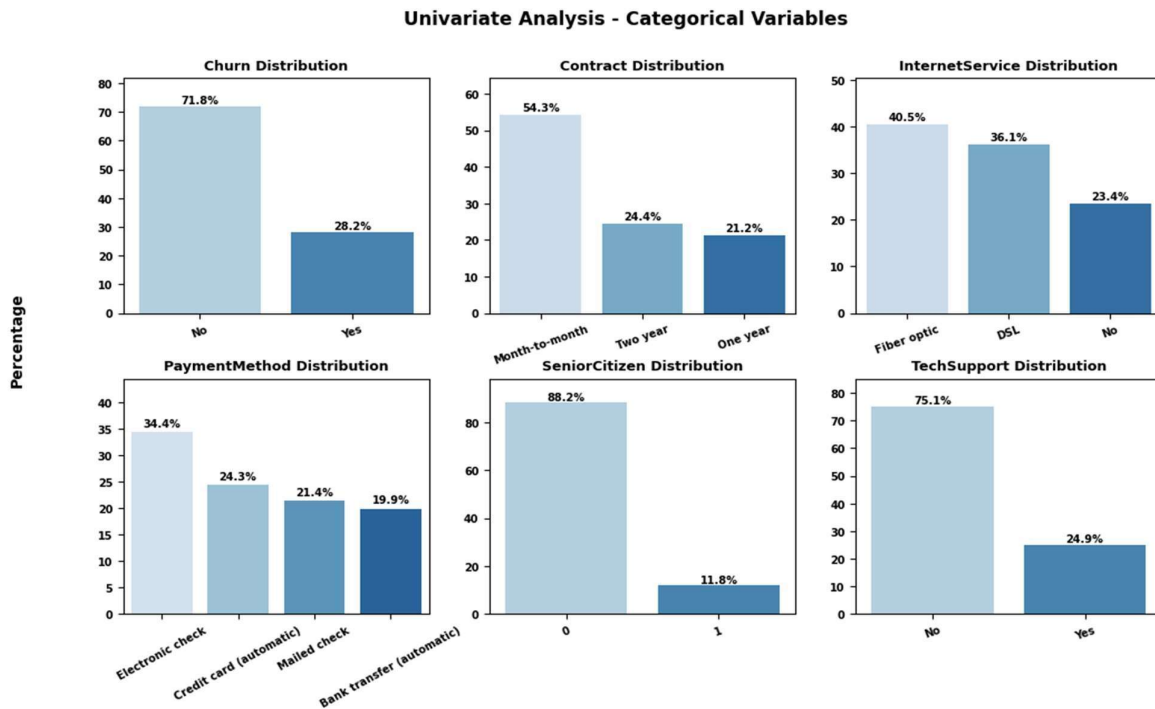


Fig 2.5.2 – Distribution of Categorical Variables

Variable	Observation	Business Insight
Churn	28.2% customers churned	Customer retention remains a key challenge
Contract	Month-to-month contracts dominate	Short-term contracts show higher churn risk
InternetService	Fiber optic has highest adoption	Premium services may influence churn
PaymentMethod	Electronic check is most common	Manual payment methods show higher churn exposure
SeniorCitizen	Senior citizens form a small segment	Seniors may require specialized support
TechSupport	Majority lack tech support	Tech support may improve customer retention

Table 2.5.2 - Univariate analysis – Categorical Variables

2.6 Bivariate Analysis

2.6.1 Numerical Features vs Churn

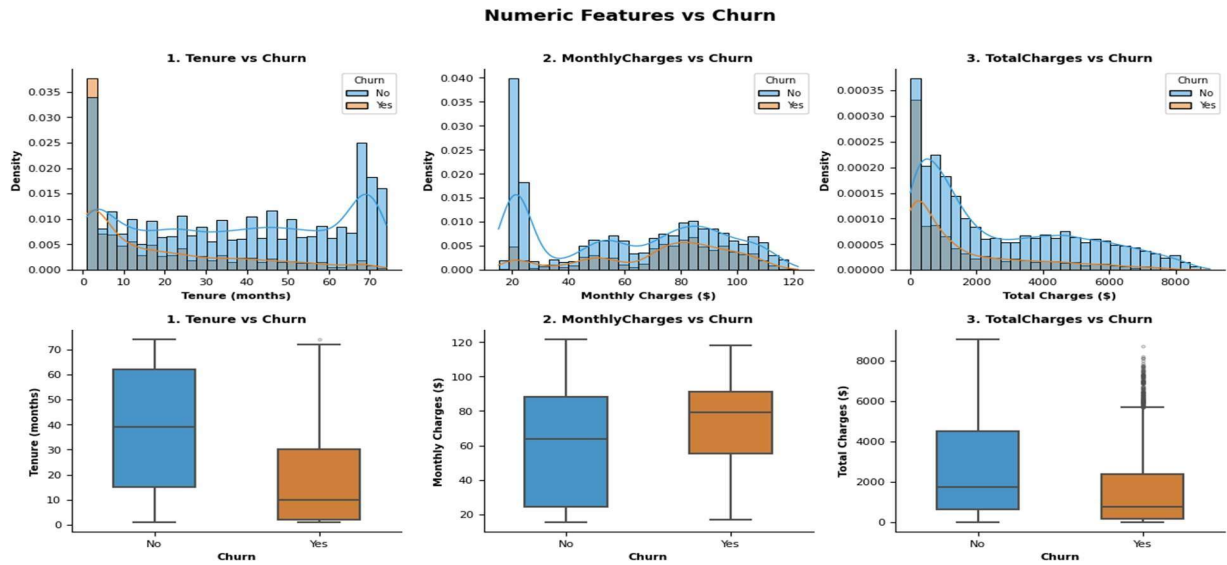


Fig 2.6.1 - Bivariate Analysis - Numerical Features vs Churn

Feature	No Churn	Churned	Key Insight
Tenure	Median ~40 months	Median ~10 months	Early-stage customers show higher churn risk
MonthlyCharges	Lower monthly charges dominate	Higher monthly charges dominate	Higher monthly charges are associated with churn
TotalCharges	Higher lifetime spends	Lower lifetime spends	Churn is concentrated among lower lifetime value customers

Table 2.6.1 - Bivariate Analysis - Numerical Features vs Churn

2.6.2 Categorical Features vs Churn

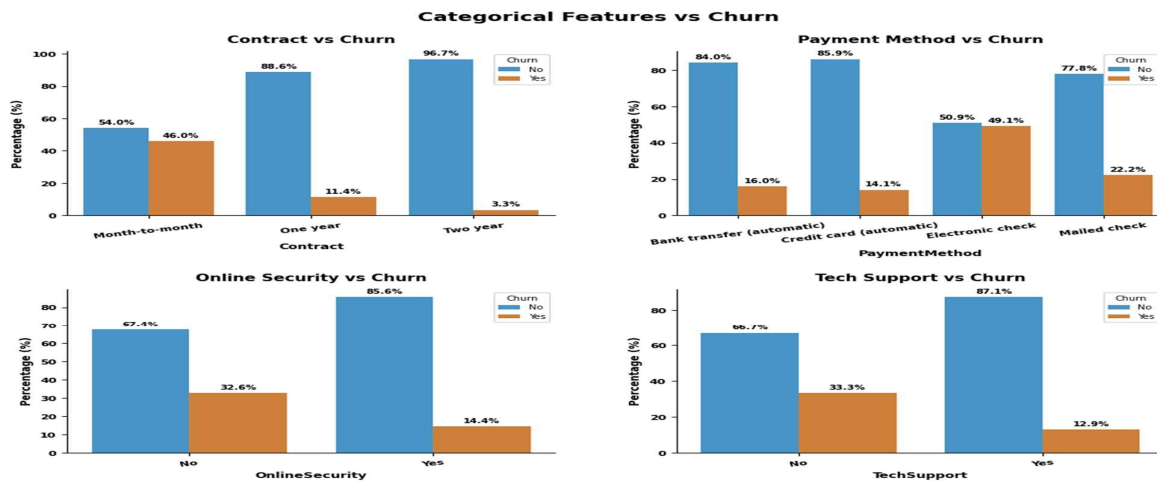


Fig 2.6.2 - Bivariate Analysis - Categorical Features vs Churn

Feature	Observation	Business Insight
Contract	Month-to-month customers show highest churn	Long-term contracts improve retention
PaymentMethod	Electronic check users show highest churn	Automatic payments reduce churn risk
OnlineSecurity	Customers without online security churn more	Security services improve retention
TechSupport	Customers without tech support churn more	Technical support lowers churn risk

Table 2.6.2 - Bivariate Analysis - Categorical Features vs Churn

- Additional categorical feature-wise churn analyses are included in Appendix B.

2.7 Multivariate Analysis

2.7.1 Heatmap

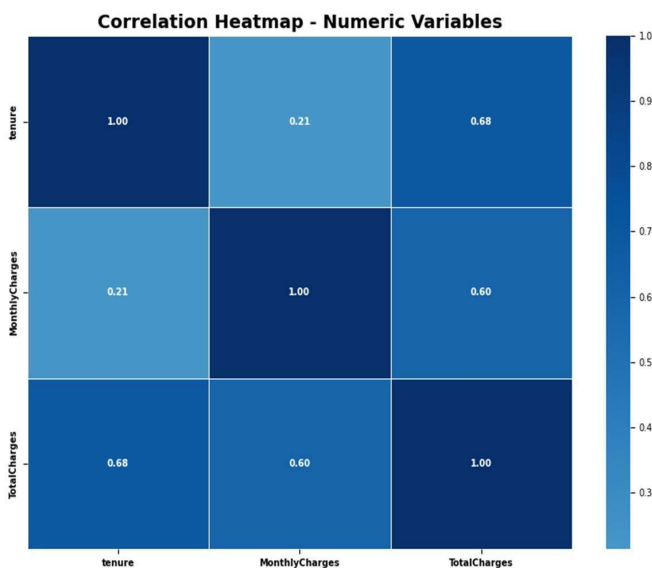


Fig 2.7.1 – Correlation Heatmap

Observation	Business & Modelling Insight
tenure and TotalCharges show moderate positive correlation (0.68)	Long-term customers contribute higher cumulative revenue
MonthlyCharges and TotalCharges show moderate correlation (0.60)	Higher-priced plans contribute greater revenue
No extremely high correlations observed	Severe multicollinearity is not observed among numerical features
tenure & MonthlyCharges show weak correlation (0.21)	Customer tenure is largely independent of pricing

Table 2.7.1 – Correlation Analysis

2.7.2 MonthlyCharges by Contract Type and Churn

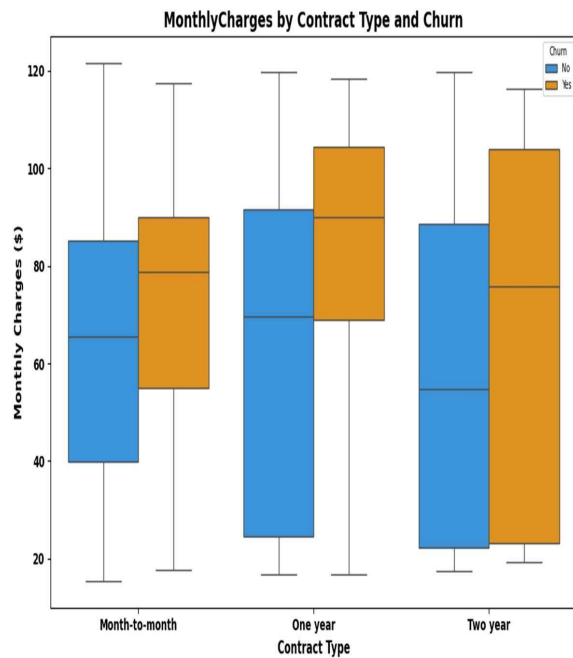


Fig 2.7.2 – Contract-wise MonthlyCharges Analysis

Contract Type	Observation	Key Insight
Month-to-Month	Churned customers show higher monthly charges	High pricing increases churn risk for short-term customers
One year	Churn is concentrated among higher-priced plans	Pricing influences churn even in annual contracts
Two year	Churn is comparatively lower overall	Long-term contracts improve customer retention

Table 2.7.2 – Contract wise MonthlyCharges Analysis

2.7.3 Tenure by Internet Service & Churn

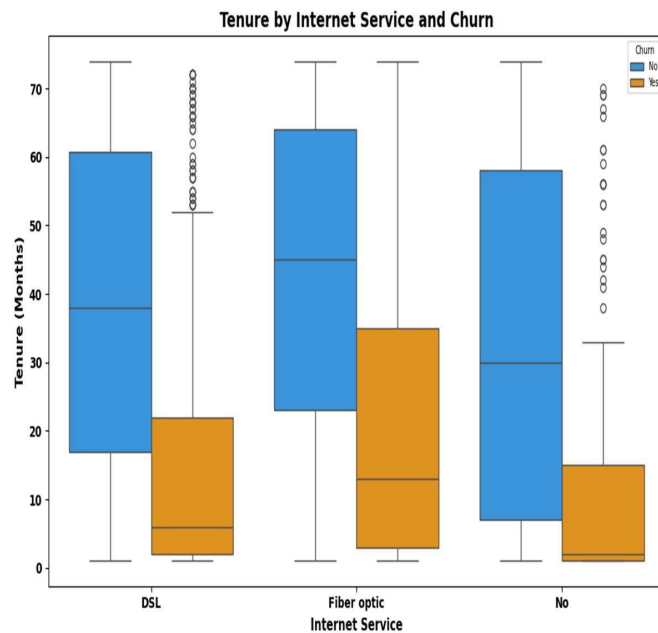


Fig 2.7.3 – Tenure by Internet Service and Churn

Internet Service	Observation	Key Insight
DSL	Churned customers generally have lower tenure	New DSL customers are more likely to churn
Fiber optic	Churned customers are concentrated at low tenure levels	Fiber optic customers show higher early-stage churn risk
No Internet Service	Churned customers mostly have very low tenure	Customers without internet service exhibit lower retention

Table 2.7.3 – Tenure Analysis by Internet Service

2.7.4 Contract Type, Internet Service, and Churn

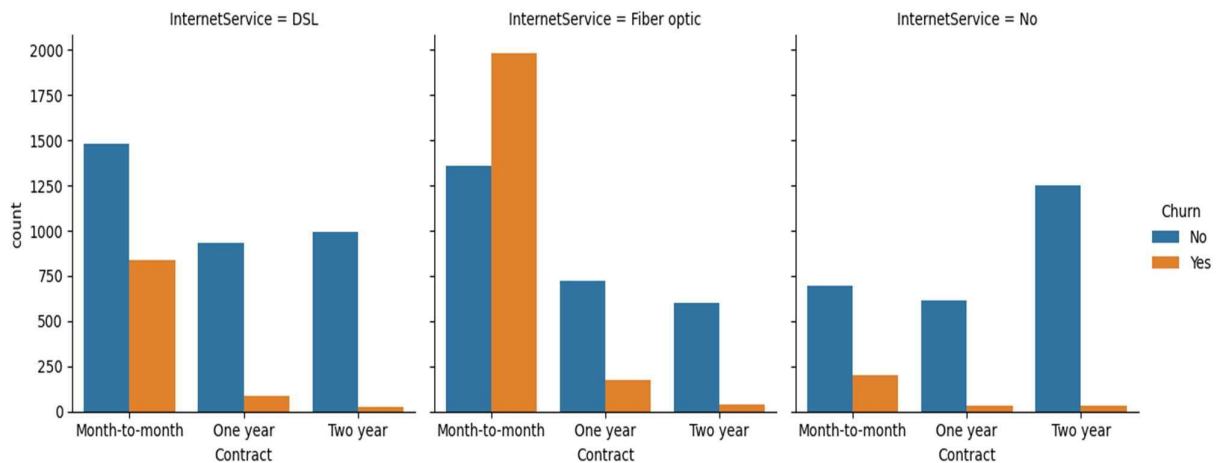


Fig 2.7.4 - Contract Type and Churn Across Internet Services

Internet Service & Contract	Observation	Key Insight
DSL + Month-to-month	Higher churn compared to long-term contracts	Short-term DSL customers are more likely to churn
Fiber optic + Month-to-month	Highest churn among all customer groups	Fiber optic users with short-term contracts are the highest-risk customers
One-year and Two-year contracts	Churn remains consistently low across internet service types	Long-term contracts strongly improve customer retention

Table 2.7.4 - Contract and Internet Service Analysis

3. Data Preprocessing

3.1 Data Cleaning Duplicate/ Anomalous Check

Step	Issue Found	Treatment Applied
Currency Symbol Removal	£ and \$ symbols in MonthlyCharges and TotalCharges	Removed symbols and converted to numeric
Duplicate Check	Duplicate records identified	Removed duplicate rows
Invalid Values	Negative and zero values in tenure and TotalCharges	Converted to NaN
Categorical Anomalies	Inconsistent categories in PaymentMethod	Standardized category labels
Target Anomalies	Multiple formats in Churn labels	Standardized category labels
Logical Inconsistency (Internet)	Internet add-ons present for customers without internet service	Corrected inconsistencies
Logical Inconsistency (Phone)	MultipleLines assigned to customers without phone service	Corrected conflicting values
Post-cleaning Duplicate Check	New duplicates created after standardization	Removed duplicate rows

Table 3.1 – Data Cleaning

3.2 Feature Selection

The Gender feature was removed prior to modelling due to its limited association with customer churn and low predictive contribution.

3.3 Train/Validation/Test Split

The dataset was split into training, validation, and test sets using stratified sampling to preserve the original churn class distribution across all subsets.

Split	Records	Percentage	Churn (Yes)	Churn (No)
Training Set	8419	70	28.22	71.78
Validation Set	1804	15	28.21	71.78
Test Set	1805	15	28.2	71.8

Table 3.3 - Train/Validation/Test Split

3.4 Missing Value Treatment

Missing value imputation was performed after data splitting using statistics derived only from the training set to prevent data leakage.

Column	Skewness	Imputation Method	Technique Used
tenure	0.22 (approximately symmetric)	Mean Imputation	SimpleImputer
MonthlyCharges	-0.22 (approximately symmetric)	Mean Imputation	SimpleImputer
TotalCharges	0.90 (right-skewed)	Median Imputation	SimpleImputer

Table 3.4 - Missing Value Treatment

Imputation methods were selected based on the skewness characteristics of numerical variables.

3.5 Outlier check

Boxplots were used to identify potential outliers in numerical variables. Mild outliers were observed primarily in the TotalCharges variable, while tenure and MonthlyCharges showed no significant extreme values. Since they represented genuine high-value customers rather than data errors, no treatment was applied.

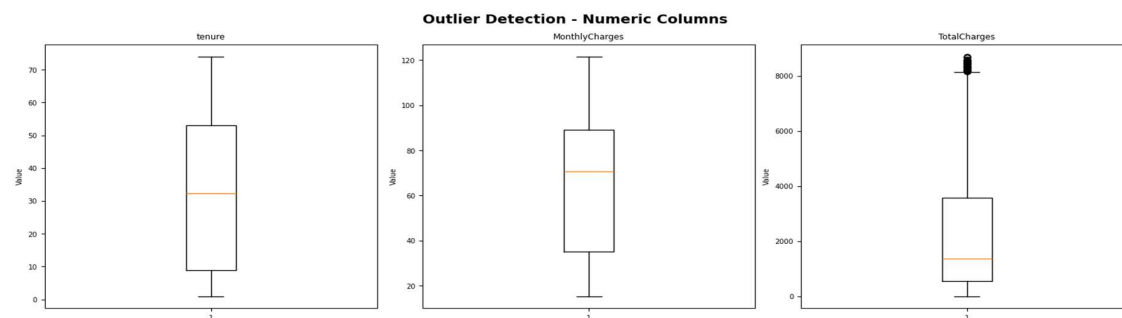


Fig 3.5 - Outlier check

3.6 Feature engineering

Categorical variables were transformed using OneHotEncoder to convert categories into machine-readable numerical representations. The encoded dataset contained 22 features across training, validation, and test sets.

Dataset	Rows	Columns After Encoding
Training Set	8419	22
Validation Set	1804	22
Test Set	1805	22

Table 3.6 – One Hot Encoding

A square root transformation was applied to the 'TotalCharges' variable to reduce skewness and improve the distribution of the feature for modelling.

3.7 Class Imbalance Handling

SMOTE was applied only on the training set to improve minority class learning while preventing data leakage.

3.8 Feature Scaling

Numerical variables were standardized using StandardScaler to ensure all features contributed proportionally during model training.

3.9 Data Leakage Handling

- Missing value imputation performed after train-test split.
- Encoding and scaling fitted only on training data.
- SMOTE applied exclusively on training data.
- Validation and test sets remained unseen during preprocessing and training.

The pre-processed dataset was subsequently used for baseline churn prediction modelling.

4. Model Building - Baseline Model

4.1 Metric of Choice and Business Rationale

The model can make two types of errors:

False Negatives (FN): Predicting a customer will stay, but they churn. This causes direct revenue and customer loss.

False Positives (FP): Predicting a customer will churn, but they stay. This incurs unnecessary marketing and retention costs.

Primary Metric Selection:

Missing a churning customer (FN) is far more costly than sending an unnecessary offer to a loyal customer (FP). Therefore, Recall is chosen as the primary metric to maximize the identification of actual churn risks and minimize customer loss.

4.2 Secondary Evaluation Metrics

Metric	Business & Technical Purpose
Accuracy	Measures overall prediction correctness.
Precision	Minimizes wasted retention costs by tracking actual churners among predicted churners.
F1-Score	Provides a harmonic balance between Precision and Recall.
Confusion Matrix	Visualizes exact correct vs. incorrect predictions across both classes.
ROC-AUC	Evaluates how well the model distinguishes between churners and non-churners.

Table 4.2 – Secondary Evaluation Metrics

4.3 Model Building

A Logistic Regression model was developed as the baseline classifier for predicting customer churn. Logistic Regression was selected due to its simplicity, interpretability, and effectiveness in binary classification problems. The model establishes a benchmark performance against which more advanced models can later be compared.

Step 1 — Initial Logistic Regression Model

An initial Logistic Regression model was developed using all pre-processed features to establish a baseline performance benchmark.

Step 2 — Handling Class Imbalance

Since the target variable exhibited class imbalance, SMOTE (Synthetic Minority Oversampling Technique) was applied to the training dataset to improve the model's ability to learn minority churn patterns and reduce false negatives.

Step 3 — Final Logistic Regression Model

A refined Logistic Regression model was trained using the balanced dataset and evaluated using multiple classification metrics including Accuracy, Precision, Recall, F1-Score, ROC-AUC, and Confusion Matrix.

Additional statistical refinement techniques such as multicollinearity assessment and p-value based feature selection may be explored in future iterations to further improve model interpretability and feature optimization. Detailed Logistic Regression coefficient summaries and statistical outputs are provided in Appendix C.

Test set evaluation was reserved exclusively for the final selected models to prevent data leakage and ensure unbiased performance reporting.

4.4 Model Comparison Table

The SMOTE-based Logistic Regression model achieved the highest Recall score (78%), making it the most suitable model for churn prediction, where minimizing False Negatives is a business priority. Although Accuracy and Precision decreased slightly, the model improved identification of potential churn customers.

Train Performance Comparison					
Model	Accuracy	Precision	Recall	F1 Score	AUC-ROC
Baseline LR	80.32	67.63	58.04	62.47	84.95
Smote LR	78.2	76.22	81.98	78.99	86.02
Validation Performance Comparison					
Model	Accuracy	Precision	Recall	F1 Score	AUC-ROC
Baseline LR	80.54	69.17	55.99	61.89	85.34
Smote LR	77.55	57.45	78.78	66.45	85.03

Table 4.3 – Baseline Model Comparison Table

The baseline Logistic Regression model achieved higher Accuracy (80.32% and 80.54) and Precision (67.63% and 69.17%), indicating better overall correctness and fewer false positive churn predictions. However, its Recall was relatively low (58.04% and 55.99%), meaning many actual churn customers were missed.

After applying SMOTE, the model's Recall improved significantly to 81.98% in train and 78.78% in validation reducing False Negatives and improving churn detection capability. Although Accuracy and Precision decreased slightly, the SMOTE-based model achieved a better F1-Score, indicating a more balanced performance between Precision and Recall. Both models demonstrated strong discriminatory ability with ROC-AUC values above 84%.

Additional evaluation outputs, including confusion matrices and detailed model performance summaries, are included in Appendix C.

4.5 Final Model Selection

SMOTE LR is selected as the better model based on the primary metric Recall. Although the baseline Logistic Regression model achieved higher Precision and Accuracy, the SMOTE Logistic Regression model significantly improved Recall and F1-score while maintaining similar ROC-AUC performance. Since minimizing missed churn customers was the primary business objective, the SMOTE Logistic Regression model was preferred.

5. Model Building - Advanced Models

After establishing the baseline Logistic Regression model, advanced machine learning models such as Decision Tree, Random Forest, AdaBoost, Gradient Boosting, and XGBoost were developed to improve churn prediction performance. In addition, model improvement techniques including multicollinearity analysis using VIF and removal of statistically insignificant variables based on p-values were applied to the Logistic Regression model to improve interpretability and overall performance.

5.1 Logistic - Model Performance Improvement

Step 1 - Detecting and Dealing with Multicollinearity

Variation Inflation Factor (VIF) method was used for testing for multicollinearity. Most features had VIF values below 5, indicating low multicollinearity. Although InternetService_Fiber optic and MonthlyCharges showed moderately higher VIF values, they were within an acceptable range and retained due to their business importance.

Step 2 - Removing high p-value variables

'Partner_Yes', 'StreamingTV_Yes', 'StreamingMovies_Yes' variables had a p-value greater than 0.05 and were dropped. We build a model with 19 features.

Step 3 – Smote on Improved LR

Smote is applied on the reduced 19 features and a model is built upon it. Detailed performance of the 'Smote Improved Lr' is provided in Appendix D.

5.2 Model Building - Original Data

Five advanced machine learning models — Decision Tree, Random Forest, AdaBoost, Gradient Boosting, and XGBoost — were built using the original dataset.

5.3 Model Building – Oversampled Data

To address class imbalance, SMOTE (Synthetic Minority Oversampling Technique) was applied to the training data to generate synthetic minority class samples and balance the dataset. The same five models were then developed and evaluated on the oversampled data.

5.4 Model Building – Undersampled Data

To address class imbalance, Random UnderSampling was applied to the training data by reducing majority class observations. The same five models were then developed and evaluated on the undersampled data.

The training and validation performance of all models is provided in Appendix E.

6. Model Performance Improvement using Hyperparameter Tuning

Hyperparameter tuning was performed to improve model generalization and enhance churn prediction capability. Since the dataset was imbalanced and minimizing false negatives was a key business objective, Recall was considered the primary evaluation metric, while ROC-AUC was used as the secondary metric to evaluate overall class discrimination performance.

GridSearchCV with 5-fold cross-validation was used to optimize model parameters. In addition, threshold tuning was applied to selected models to identify optimal probability cutoffs and improve Recall while maintaining reasonable Precision and overall model stability.

Although XGBoost showed signs of overfitting in the initial evaluation, it was still included for tuning because boosting-based models often improve significantly after hyperparameter optimization and threshold adjustment. Based on validation performance, generalization capability, Recall, and ROC-AUC scores, the following models were shortlisted for tuning:

- Gradient Boosting + SMOTE
- AdaBoost + Undersampling
- XGBoost + Original Data
- AdaBoost + SMOTE
- Gradient Boosting + Undersampling

Detailed hyperparameter configurations and train-validation performance analyses are provided in Appendix F.

7. Model Comparison and Final Model Selection

A total of eight tuned model variations were evaluated, including both default and threshold-optimized versions. Based on Recall, ROC-AUC performance, and train-validation generalization capability, five final tuned models were shortlisted for final comparison and model selection.

- AdaBoost Undersampled (Threshold: 0.51)
- XGBoost Original (Threshold: 0.55)
- AdaBoost SMOTE (Threshold: 0.50)
- GBM Undersampled (Threshold: 0.55)
- GBM SMOTE (Threshold: 0.50)

The model training performance analysis is provided in Appendix G.

7.1 Tuned Models – Validation Performance

--- Validation Performance Comparison (Top 5 Tuned Models) ---					
	Accuracy	Recall	Precision	F1	AUC-ROC
AdaBoost Undersampled (0.51)	77.77%	78.98%	57.76%	66.72%	85.35%
XGBoost Original (0.55)	79.55%	74.85%	61.25%	67.37%	85.59%
AdaBoost SMOTE (0.50)	78.33%	77.80%	58.75%	66.95%	85.15%
GBM Undersampled (0.55)	78.88%	75.83%	59.94%	66.96%	85.35%
GBM SMOTE (0.50)	79.05%	73.28%	60.65%	66.37%	84.96%

Table 7.1 Tuned Models – Validation Performance

- AdaBoost + Undersample achieved the highest Recall (78.98%), demonstrating strong churn detection capability.
- XGBoost + Original achieved the highest F1-score (67.37%) and ROC-AUC (85.59%), indicating more balanced and stable classification performance.
- GBM-based models demonstrated comparatively balanced Precision, Recall, and F1-score performance across validation data.
- Overall, all five tuned models showed stable validation performance with relatively close metric values, indicating good generalization capability.

7.2 Tuned Models – Test Performance

--- Test Performance Comparison (Top 5 Tuned Models) ---					
	Accuracy	Recall	Precision	F1	AUC-ROC
AdaBoost Undersampled (0.51)	75.79%	78.00%	54.99%	64.50%	84.50%
XGBoost Original (0.55)	77.51%	72.30%	58.14%	64.45%	84.76%
AdaBoost SMOTE (0.50)	76.18%	74.85%	55.78%	63.93%	84.57%
GBM Undersampled (0.55)	76.73%	73.28%	56.77%	63.98%	84.65%
GBM SMOTE (0.50)	76.95%	72.10%	57.25%	63.83%	84.34%

Table 7.2 Tuned Models – Test Performance

- AdaBoost + Undersample achieved the highest Recall (78.00%), demonstrating the strongest churn detection capability among the tuned models.
- XGBoost + Original achieved the highest Precision (58.14%) and ROC-AUC (84.76%), indicating more balanced and stable classification performance.
- GBM-based models demonstrated comparatively stable performance across Recall, Precision, and F1-score without large metric fluctuations.
- Although AdaBoost + Undersample achieved superior Recall, XGBoost + Original demonstrated better overall balance between Recall, Precision, F1-score, and ROC-AUC, making it a more generalized and stable model.

7.3 Final Model Selection

Among the tuned models, AdaBoost + Undersample with threshold optimization (0.51) was identified as the best-performing model because it achieved the highest Recall (78.00%) along with a strong F1-score (64.50%). This indicates superior capability in identifying potential churn customers while maintaining balanced overall classification performance.

XGBoost + Original Data with threshold optimization (0.55) was selected as the second-best model due to its more consistent and balanced performance across all evaluation metrics. The model achieved the highest Precision (58.14%) and ROC-AUC (84.76%) among the shortlisted models, indicating stronger class discrimination capability and better overall generalization performance.

--- AdaBoost Undersampled (Threshold: 0.51) ---					
	Accuracy	Recall	Precision	F1	AUC-ROC
Train	76.21%	80.22%	55.42%	65.55%	84.98%
Validation	77.77%	78.98%	57.76%	66.72%	85.35%
Test	75.79%	78.00%	54.99%	64.50%	84.50%
--- XGBoost Original (Threshold: 0.55) ---					
	Accuracy	Recall	Precision	F1	AUC-ROC
Train	79.94%	78.45%	61.30%	68.82%	88.13%
Validation	79.55%	74.85%	61.25%	67.37%	85.59%
Test	77.51%	72.30%	58.14%	64.45%	84.76%

Table 7.3 – Final model Performances

AdaBoost + Undersample demonstrated superior churn detection capability with the highest Recall across all datasets, while XGBoost + Original showed more balanced and consistent performance across Precision, F1-score, and ROC-AUC, indicating stronger overall generalization capability. Supporting evaluation visualizations, including confusion matrices, ROC-AUC curves, and Precision-Recall curves for the final shortlisted models, are provided in Appendix G.

7.4 Key Feature Importance

Feature	XGBoost Rank	AdaBoost Rank	Business Significance
Contract_Month-to-month	1st (0.60)	3rd	Strongest churn driver
tenure	8th	1st (0.21)	New customers churn most
MonthlyCharges	Not in top 10	2nd (0.17)	High charges drive churn
TotalCharges	10th	4th	Cumulative spend indicator
TechSupport_Yes	5th	5th	Lack of support increases churn
PaymentMethod_Electronic check	3rd	6th	Highest churn payment method
InternetService_Fiber optic	2nd	7th	Premium service, high churn
Dependents_Yes	9th	8th	Family customers more stable
Contract_One year	6th	9th	Medium-term contracts reduce churn
OnlineSecurity_Yes	7th	10th	Security service reduces churn

Table 7.4 – Top Features of Final Models

Both final models identified contract type, tenure, monthly charges, internet service type, and technical support as important drivers of customer churn. Month-to-month contracts and fiber optic internet service were consistently associated with higher churn risk across the models. More feature importance is provided in Appendix G.

8. Actionable Insights & Recommendations

Priority	Insight	Recommendation	Target Segment
HIGH	Month-to-month contracts drive highest churn	Offer discounts to switch to annual contracts — 'Lock-in & Save' campaign	All month-to-month customers
HIGH	New customers churn most (low tenure)	Implement structured 6-month onboarding program with loyalty rewards at 3 and 6 months	Tenure < 6 months
MEDIUM	High monthly charges increase churn	Tiered loyalty discounts for customers paying above \$75/month for 6+ months	MonthlyCharges > \$75
MEDIUM	Fiber optic customers churn more despite premium service	Service quality audit + proactive satisfaction surveys + price locks	Fiber optic customers
MEDIUM	Lack of tech support increases churn	Bundle complimentary tech support trials for at-risk customers	Customers without tech support
LOW	Electronic check users churn at 49.1%	Incentivize switch to automatic payments with bill discounts	Electronic check users
LOW	Senior citizens churn at 42.4% vs 26.3%	Dedicated senior retention program with simplified billing and special pricing	Senior customers
LOW	No online security correlates with higher churn	Offer free online security trials to high-risk customers	Customers without security

Table 8.1 - Actionable Insights & Recommendations

9. Conclusion

Two complementary models were developed to address AlphaCom's churn problem:

- AdaBoost Undersampled (Threshold: 0.51) maximizes churn detection with a recall of 78%, ensuring most at-risk customers are identified proactively.
- XGBoost Original (Threshold: 0.55) provides the most balanced and reliable predictions with superior precision (58.14%), accuracy (77.51%), and AUC-ROC (0.848).

Both models significantly outperform the random classifier baseline and the initial Logistic Regression model. Together, they enable AlphaCom to proactively identify at-risk customers, design targeted retention strategies, and ultimately safeguard revenue and strengthen competitive position.

Contract type, customer tenure, and monthly charges were identified as the top churn drivers, providing clear direction for targeted retention campaigns.

Appendix A – Data Dictionary

Appendix A.1 Detailed Data Dictionary

Feature	Type	Description
Gender	Categorical	Customer gender (Male/Female)
SeniorCitizen	Binary	Whether customer is senior (0=No, 1=Yes)
Partner	Categorical	Indicates whether the customer has a partner.
Dependents	Categorical	Specifies whether the customer has dependents.
Tenure	Numeric	Number of months with company
PhoneService	Categorical	Phone service status
MultipleLines	Categorical	Multiple lines status
InternetService	Categorical	Type of internet service subscribed (DSL, Fiber Optic).
OnlineSecurity	Categorical	Online security subscription status
OnlineBackup	Categorical	Online backup status
DeviceProtection	Categorical	Device protection status
TechSupport	Categorical	Tech support status
StreamingTV	Categorical	TV streaming status
StreamingMovies	Categorical	Movie streaming status
Contract	Categorical	Customer contract type (month-to-month, one-year, or two-year)
PaperlessBilling	Categorical	Paperless billing enrolment
PaymentMethod	Categorical	Customer payment method
MonthlyCharges	Numeric	Monthly amount charged
TotalCharges	Numeric	Total amount charged
Churn	Target Variable	Customer churn status

Table A.1 - Detailed Data Dictionary

Appendix B – Additional Categorical Analyses

Appendix B.1 Additional Categorical Features vs Churn

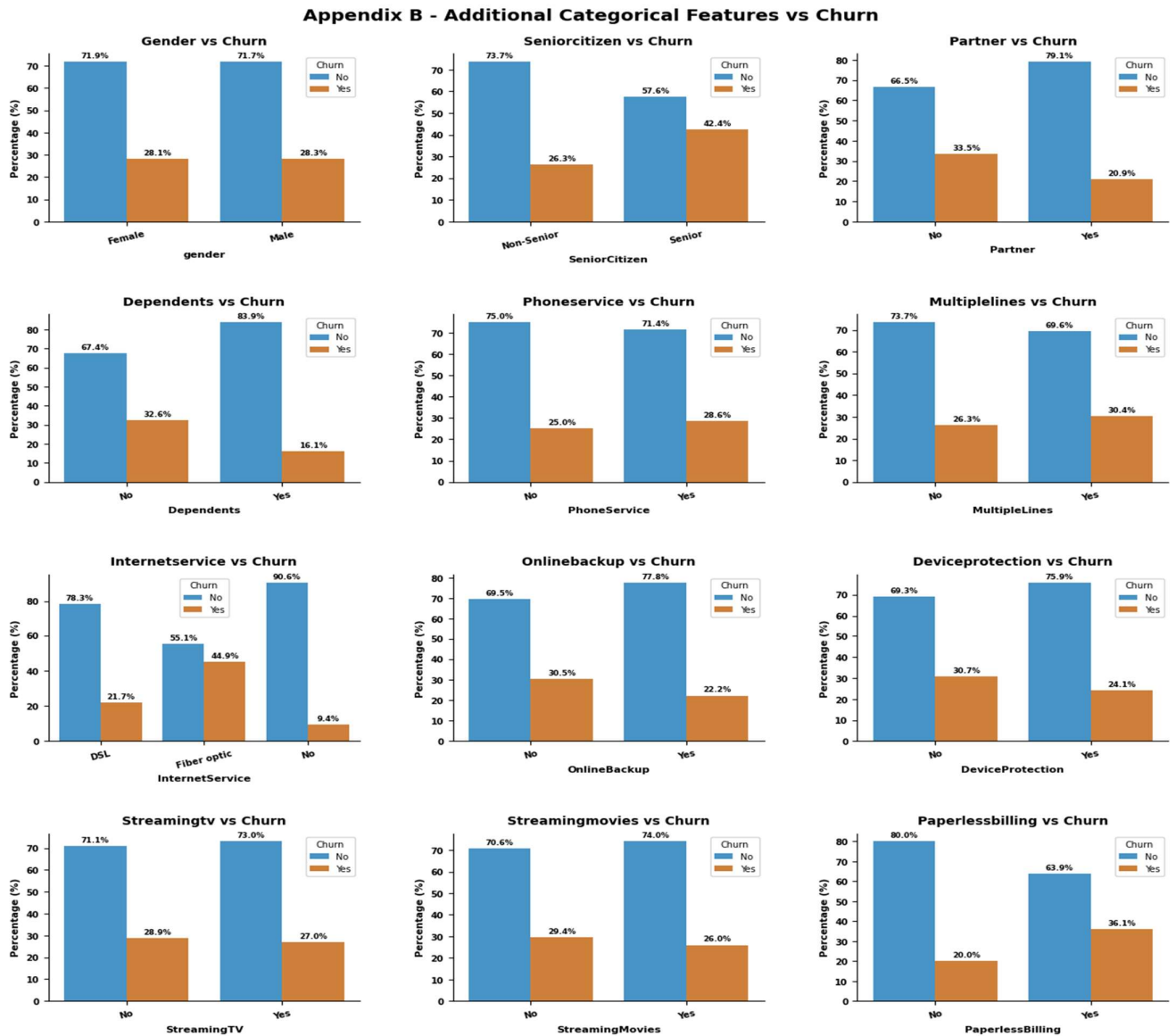


Fig B.1 – Additional Categorical Features vs Churn

- Customers lacking additional service subscriptions generally show higher churn tendencies, indicating the importance of service engagement for retention.

Appendix B.2 Churn Rate Heatmap

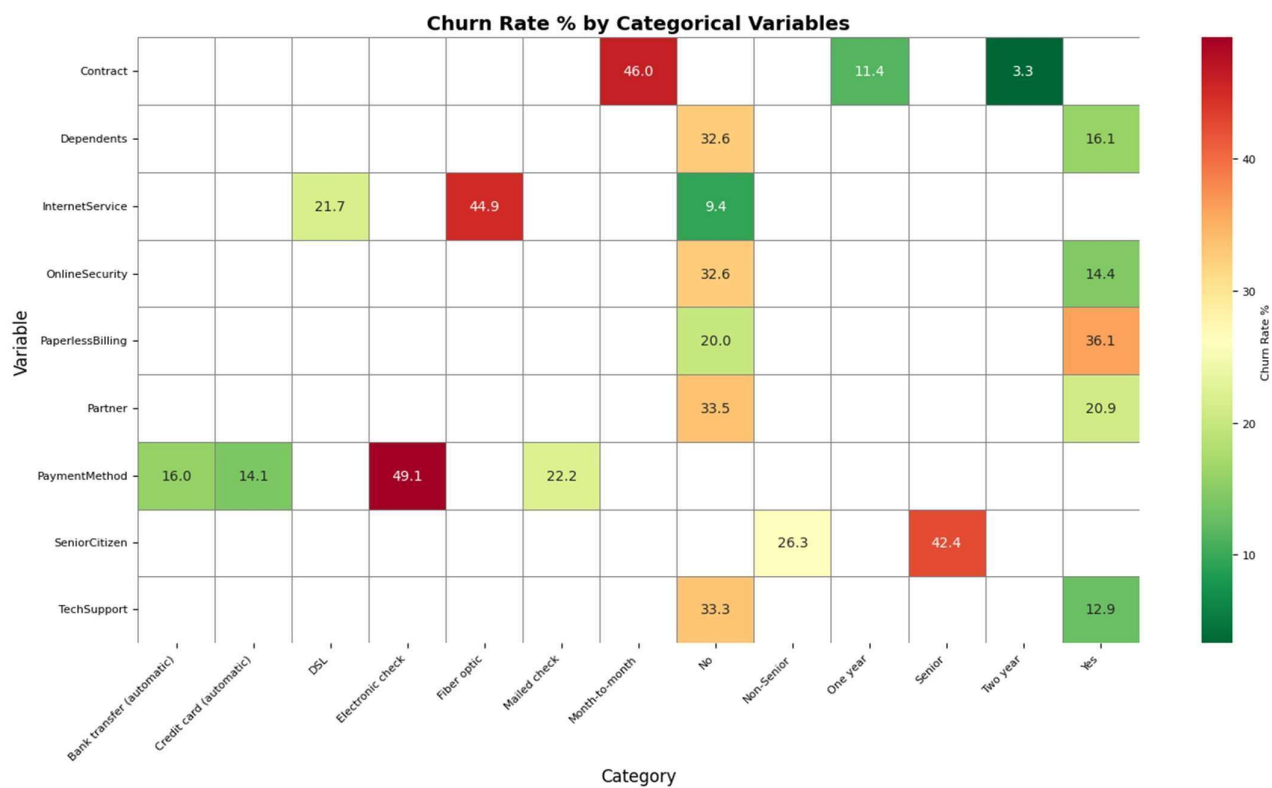


Fig B.2 – Churn Rate Across Key Categorical Variables

- Month-to-month contracts, Fiber optic services, electronic check payments, and absence of support/security services show comparatively higher churn rates.

Appendix C – Additional Model Evaluation Outputs

Appendix C.1 Logistic Regression Summary Table

Feature	Coefficient	p-value	Interpretation
tenure	-0.3386	<0.001	Customers with longer tenure are less likely to churn.
MonthlyCharges	0.2591	0.003	Higher monthly charges are associated with a higher likelihood of churn.
TotalCharges	-0.3639	<0.001	Customers with higher cumulative spending are less likely to churn, indicating stronger retention.
Dependents_Yes	-0.3777	<0.001	Customers with dependents are less likely to churn than those without dependents.
OnlineSecurity_Yes	-0.4389	<0.001	Customers subscribed to online security services are less likely to churn.
OnlineBackup_Yes	-0.2846	<0.001	Customers with online backup services are less likely to churn.
TechSupport_Yes	-0.6434	<0.001	Customers with technical support are significantly less likely to churn.
InternetService_Fiber optic	1.4747	<0.001	Customers using fiber optic internet are more likely to churn than DSL users.
Contract_One year	0.6822	<0.001	Compared to two-year contracts (reference category), one-year contract customers are more likely to churn.
Contract_Month-to-month	1.4956	<0.001	Compared to two-year contracts (reference category), month-to-month customers are much more likely to churn.
PaymentMethod_Electronic check	0.3477	<0.001	Customers using electronic check payments are more likely to churn than customers using the reference payment method.

Table C.1 Logistic Regression Summary Table

- Positive coefficients indicate increased churn probability, while negative coefficients indicate reduced churn likelihood.

Appendix C.2 Baseline Logistic Regression Performance

Metric	Train Set	Validation Set
Accuracy	80.32%	80.54%
Precision	67.63%	69.17%
Recall	58.04%	55.99%
F1-Score	62.47%	61.89%
ROCAUC	84.95%	85.34%

Table C.2 – Baseline Logistic Regression Performance

- The baseline Logistic Regression model demonstrated stable performance across validation and test datasets, indicating good generalization capability.

Appendix C.3 Confusion Matrix – Baseline Logistic Regression

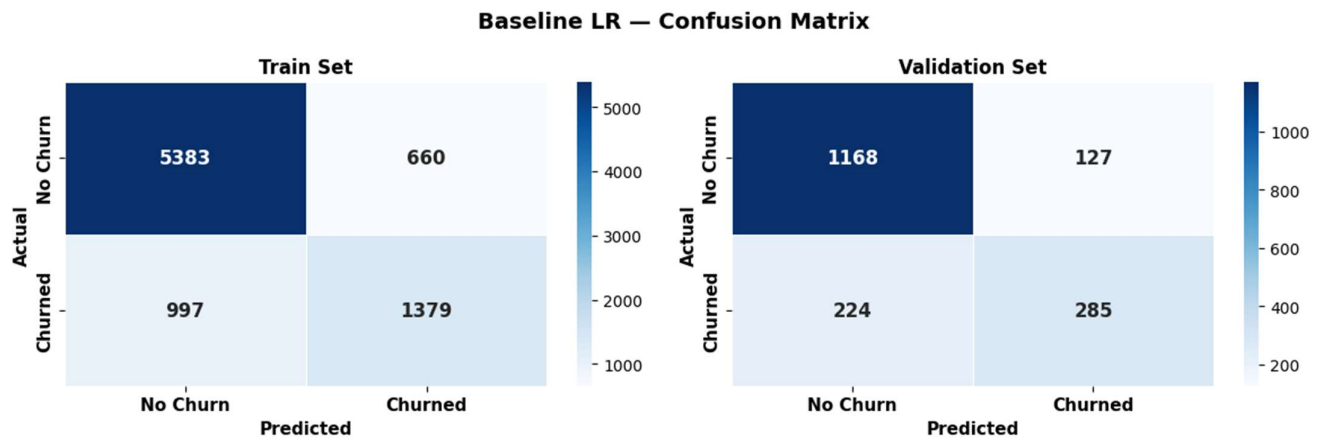


Fig C.3 - Confusion Matrix – Baseline Logistic Regression

Observations

- The model correctly classified a large proportion of non-churn customers in both the training set (5,383) and validation set (1,168), indicating strong performance in identifying customers who are likely to stay.
- However, the model produced 997 false negatives in the training set and 224 false negatives in the validation set, meaning a considerable number of actual churn customers were incorrectly classified as non-churn customers.
- Overall, the baseline Logistic Regression model achieved good performance in identifying non-churn customers but was less effective at detecting churn customers, highlighting the need for class imbalance handling techniques such as SMOTE or undersampling to improve churn detection.

Appendix C.4 SMOTE Logistic Regression Performance

Metric	Train Set	Validation Set
Accuracy	78.20%	77.55%
Precision	76.22%	57.45%
Recall	81.98%	78.78%
F1-Score	78.99%	66.45%
ROCAUC	86.02%	85.03%

Table C.4 - SMOTE Logistic Regression Performance

- The SMOTE-based model significantly improved Recall performance while maintaining stable ROC-AUC capability.

Appendix C.5 Confusion Matrix –Logistic Regression(Smote Model)

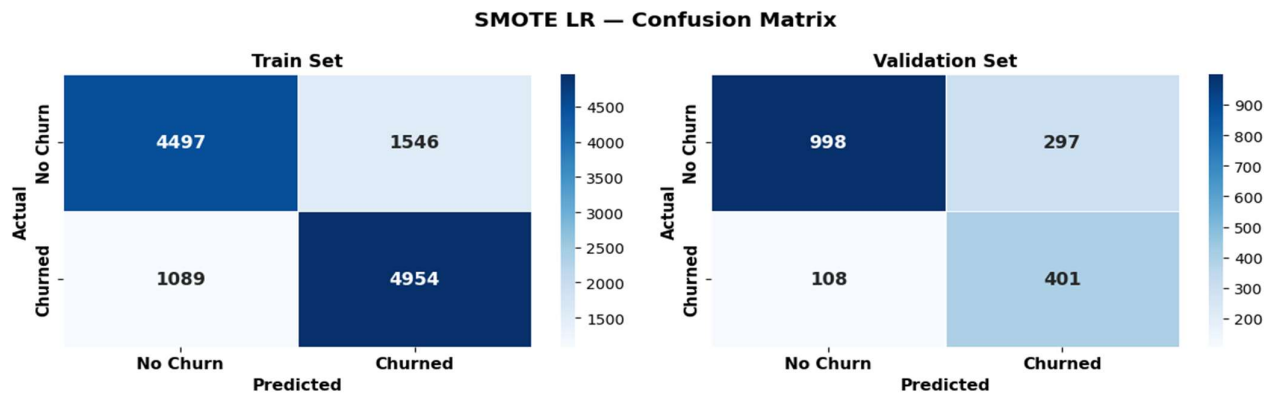


Fig C.5 - Confusion Matrix – Logistic Regression(Smote Model)

Observations – SMOTE Logistic Regression Confusion Matrix

- The number of **false negatives** decreased significantly from **224 to 108** on the validation set, indicating that the model missed fewer actual churn customers. This contributed to the improvement in **Recall from approximately 56% to 79%**.
- However, the number of **false positives** increased from **127 to 297** on the validation set, meaning more non-churn customers were incorrectly classified as churn customers.
- As a result, while Recall improved considerably, **Precision decreased**, reflecting the trade-off between identifying more churners and generating more false alarms.
- Overall, the SMOTE Logistic Regression model achieved significantly better churn detection performance than the baseline model and was more aligned with the business objective of minimizing missed churn customers.

Appendix D - Improved LR & SMOTE Improved LR Performance Analysis

Appendix D.1 Improved LR Performance

Improved LR					
	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Train	80.39%	67.76%	58.21%	62.62%	84.95%
Validation	80.27%	68.52%	55.60%	61.39%	85.33%

Table D.1 Improved LR Performance

- After removing statistically insignificant variables, the final Logistic Regression model maintained stable performance across both training and validation datasets, indicating good generalization ability.
- The model achieved strong Accuracy ~80% and ROC-AUC ~85%, showing good overall prediction capability and class separation performance.
- Precision remained higher than Recall, indicating that the model is relatively conservative in predicting churn customers and generates fewer false positives.
- The comparatively lower Recall suggests that some churn customers are still being missed by the model.
- The minimal difference between training and validation metrics indicates that the model does not suffer from severe overfitting.

Appendix D.2 Improved SMOTE LR Performance

SMOTE Improved LR					
	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Train	78.05%	76.02%	81.95%	78.87%	85.90%
Validation	77.33%	57.04%	79.57%	66.45%	85.06%

Table D.2 Improved SMOTE LR Performance

- After applying SMOTE on the improved Logistic Regression model, Recall increased significantly to ~80% on the validation set, indicating much better churn detection capability.
- The model achieved a strong F1-score (66.45%), showing a better balance between Precision and Recall compared to the baseline Logistic Regression model.
- Although Precision decreased on the validation set, the higher Recall suggests that the model is more effective in minimizing missed churn customers.
- ROC-AUC remained stable at around 85%, indicating good class separation performance even after applying SMOTE.
- The train and validation metrics are reasonably close, suggesting that the SMOTE-based improved Logistic Regression model generalizes well without severe overfitting.

Appendix E – Model Building - Advanced Models Performances

Appendix E.1 Original Data - Performance

Model	Tr. Acc	Tr. Recall	Tr. Prec	Tr. F1	Tr. AUC	Val. Acc	Val. Recall	Val. Prec	Val. F1	Val. AUC
DTree	99.86%	100.00%	99.50%	99.75%	99.90%	74.06%	50.69%	54.32%	52.44%	66.97%
RF	99.86%	100.00%	99.50%	99.75%	99.90%	79.32%	51.47%	67.53%	58.42%	70.87%
GBM	81.97%	59.55%	71.75%	65.09%	75.17%	80.93%	55.80%	70.47%	62.28%	73.30%
AdaBoost	80.19%	56.48%	67.91%	61.67%	73.00%	80.93%	55.21%	70.78%	62.03%	73.12%
XGBoost	91.55%	97.98%	77.83%	86.75%	93.50%	77.00%	67.39%	57.94%	62.31%	74.08%

Table E.1 - Original Data Performance

- Decision Tree and Random Forest achieved near-perfect performance on the training set but showed a significant decline on the validation set, indicating severe overfitting and poor generalization capability.
- XGBoost achieved the highest Recall (67.39%) and F1-score (62.31%) on the validation set, demonstrating strong churn detection capability among the advanced models.
- Gradient Boosting and AdaBoost showed more stable and generalized performance across training and validation datasets compared to tree-based models.
- Gradient Boosting achieved the highest Precision (70.47%) on the validation set, indicating better control over false positive predictions.
- Although XGBoost showed some degree of overfitting due to the large train-validation gap, it still maintained comparatively stronger Recall and balanced overall performance.
- Ensemble boosting models (Gradient Boosting, AdaBoost, and XGBoost) generalized better than standalone tree-based models, making them more suitable for churn prediction.

Appendix E.2 SMOTE Data - Performance

Model	Tr. Acc	Tr. Recall	Tr. Prec	Tr. F1	Tr. AUC	Val. Acc	Val. Recall	Val. Prec	Val. F1	Val. AUC
DTree	99.91%	99.82%	100.00%	99.91%	99.91%	72.12%	54.22%	50.55%	52.32%	66.69%
RF	99.91%	99.97%	99.85%	99.91%	99.91%	79.16%	60.12%	63.88%	61.94%	73.38%
GBM	84.83%	86.79%	83.52%	85.13%	84.83%	79.93%	69.94%	63.01%	66.29%	76.90%
AdaBoost	79.94%	84.33%	77.53%	80.79%	79.94%	77.61%	77.80%	57.64%	66.22%	77.66%
XGBoost	93.21%	93.70%	92.79%	93.24%	93.21%	77.88%	60.31%	60.91%	60.61%	72.55%

Table E.2 - SMOTE Data Performance

- Decision Tree and Random Forest achieved near-perfect performance on the oversampled training data but showed a significant decline on the validation set, indicating severe overfitting after applying SMOTE.
- XGBoost achieved very high training performance; however, validation Recall and F1-score dropped considerably, suggesting moderate overfitting and reduced generalization capability.
- Gradient Boosting and AdaBoost demonstrated comparatively more stable and generalized performance between training and validation datasets.

- AdaBoost achieved the highest Recall (77.80%) on the validation set, indicating strong capability in identifying churn customers after oversampling.
- Gradient Boosting achieved the highest F1-score (66.29%) and Accuracy (79.93%) among the oversampled models, indicating the best balance between Precision and Recall.
- Although oversampling improved Recall for most models, complex tree-based models became more prone to overfitting on the synthetic training data.
- Overall, Gradient Boosting and AdaBoost emerged as the most balanced and generalized models among the oversampled advanced models.

Appendix E.3 Undersampled Data – Performance

Model	Tr. Acc	Tr. Recall	Tr. Prec	Tr. F1	Tr. AUC	Val. Acc	Val. Recall	Val. Prec	Val. F1	Val. AUC
DTree	99.85%	99.71%	100.00%	99.85%	99.85%	69.51%	68.57%	47.23%	55.93%	69.22%
RF	99.85%	99.92%	99.79%	99.85%	99.85%	76.61%	74.66%	56.46%	64.30%	76.02%
GBM	80.18%	83.25%	78.43%	80.77%	80.18%	76.27%	78.19%	55.66%	65.03%	76.86%
AdaBoost	77.67%	82.28%	75.34%	78.66%	77.67%	75.78%	80.75%	54.80%	65.29%	77.28%
XGBoost	94.44%	96.72%	92.51%	94.57%	94.44%	75.94%	77.80%	55.23%	64.60%	76.51%

Table E.3 - Undersampled Data Performance

- Decision Tree and Random Forest achieved extremely high training performance but showed a substantial decline on the validation set, indicating severe overfitting after applying undersampling.
- XGBoost achieved very high training Recall and F1-score; however, the noticeable drop in validation performance suggests moderate overfitting and reduced generalization capability.
- Gradient Boosting and AdaBoost demonstrated comparatively better stability and generalization between training and validation datasets.
- AdaBoost achieved the highest Recall (80.75%) on the validation set, indicating strong capability in identifying churn customers after undersampling.
- AdaBoost also achieved the highest F1-score (65.29%) among the undersampled models, indicating the best balance between Precision and Recall.
- Although undersampling improved Recall across most models, reducing majority class observations also resulted in lower Accuracy and Precision in several models.
- Overall, AdaBoost and Gradient Boosting emerged as the most balanced and generalized models among the undersampled advanced models.

Appendix F - Hyperparameter Tuning

Appendix F.1 – Best Parameter values

Model	Learning Rate	Max Depth	Estimators	Min Samples Leaf	L1 Regularization (Alpha)	L2 Regularization (Lambda)
GBM + SMOTE	0.1	3	50	20	—	—
AdaBoost + UnderSample	0.5	—	150	—	—	—
XGBoost + Original	0.1	4	100	—	1	10
AdaBoost + SMOTE	1	—	150	—	—	—
GBM + UnderSample	0.1	3	100	20	—	—

Table F.1 – Best Parameter values

Appendix F.2 Default Threshold AdaBoost + Undersample Performance

--- Tuned Undersampled AdaBoost Performance Matrix ---					
	Accuracy	Recall	Precision	F1	AUC-ROC
Train	75.20%	81.36%	54.02%	64.93%	77.07%
Validation	77.00%	80.35%	56.49%	66.34%	78.01%

Table F.2 - Default Threshold AdaBoost + Undersample Performance

- The tuned AdaBoost model with undersampling achieved high Recall on both training and validation datasets, indicating strong capability in identifying churn customers.
- The train and validation metrics remained relatively close, suggesting stable generalization without severe overfitting.
- Although Precision was comparatively lower, the model maintained a balanced F1-score, making it suitable for churn prediction where minimizing missed churn customers is important.

Appendix F.3 Default Threshold Tuned XGBoost Performance

--- Tuned XGBoost Performance Matrix ---					
	Accuracy	Recall	Precision	F1	AUC-ROC
Train	78.51%	83.25%	58.37%	68.62%	79.95%
Validation	77.55%	77.80%	57.56%	66.17%	77.63%

Table F.3 – Default Threshold XGBoost Performance

- The tuned XGBoost model achieved strong Recall and F1-score performance on both training and validation datasets.
- Train and validation metrics remained reasonably close, indicating moderate generalization capability without severe overfitting after tuning.
- Although Precision was comparatively moderate, the model maintained balanced overall performance and demonstrated effective churn detection capability.

Appendix F.4 Tuned GBM + Undersample Performance

--- Tuned Undersampled GBM Performance Matrix ---					
	Accuracy	Recall	Precision	F1	AUC-ROC
Train	77.12%	82.70%	56.47%	67.11%	78.82%
Validation	76.83%	78.59%	56.42%	65.68%	77.36%

Table F.4 - Tuned GBM + Undersample Performance

- The tuned GBM + undersample model achieved strong Recall performance on both training and validation datasets, indicating effective churn detection capability.
- Train and validation metrics remained relatively close, suggesting stable generalization without severe overfitting after tuning.
- The model maintained balanced F1-score and ROC-AUC performance, demonstrating good overall classification stability after undersampling.

Appendix G – Model Comparison

Appendix G.1 Training Performance comparison

--- Training Performance Comparison (Top 5 Tuned Models) ---					
	Accuracy	Recall	Precision	F1	AUC-ROC
AdaBoost Undersampled (0.51)	76.21%	80.22%	55.42%	65.55%	84.98%
XGBoost Original (0.55)	79.94%	78.45%	61.30%	68.82%	88.13%
AdaBoost SMOTE (0.50)	77.17%	78.28%	56.95%	65.93%	85.09%
GBM Undersampled (0.55)	78.79%	78.62%	59.38%	67.66%	86.80%
GBM SMOTE (0.50)	78.71%	76.14%	59.62%	66.88%	85.70%

Table G.1 – Tuned Models - Training Performance comparison

- XGBoost achieved the highest AUC-ROC (88.13%) and F1-score (68.82%) among the tuned training models, indicating strong overall classification capability.
- AdaBoost with undersampling achieved the highest Recall (80.22%), demonstrating strong churn detection capability.
- The GBM-based models maintained comparatively balanced Precision, Recall, and F1-score performance across training data.
- Overall, all five shortlisted models demonstrated stable and competitive performance, making them suitable candidates for final model comparison.

Appendix G.2 AdaBoost + Undersample Confusion Matrices

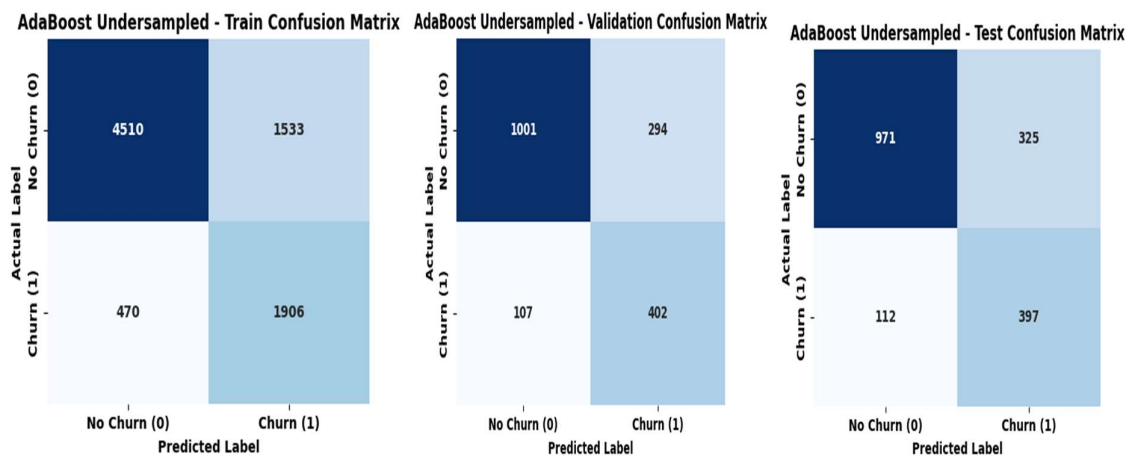


Fig G.2 - AdaBoost + Undersample Confusion Matrices

The confusion matrices indicate that the AdaBoost + Undersample model maintained strong churn detection capability across training, validation, and test datasets with relatively stable prediction patterns, supporting good generalization performance.

Appendix G.3 XGBoost + Original Confusion Matrices

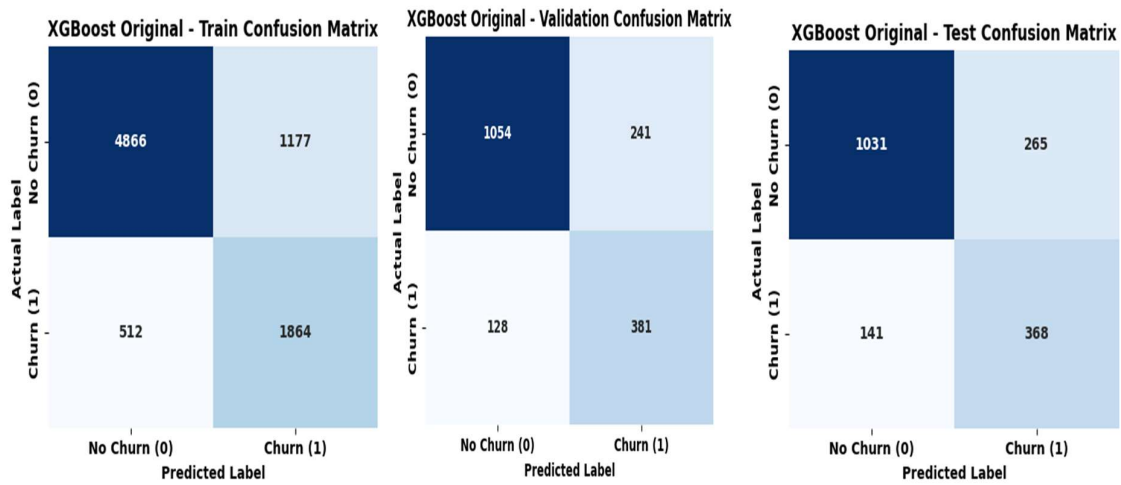


Fig G.3 - XGBoost + Original Confusion Matrices

The confusion matrices indicate that the XGBoost + Original model maintained comparatively balanced prediction performance across training, validation, and test datasets, supporting stable generalization capability and strong overall classification performance.

Appendix G.4 ROC & PR Curve Comparison

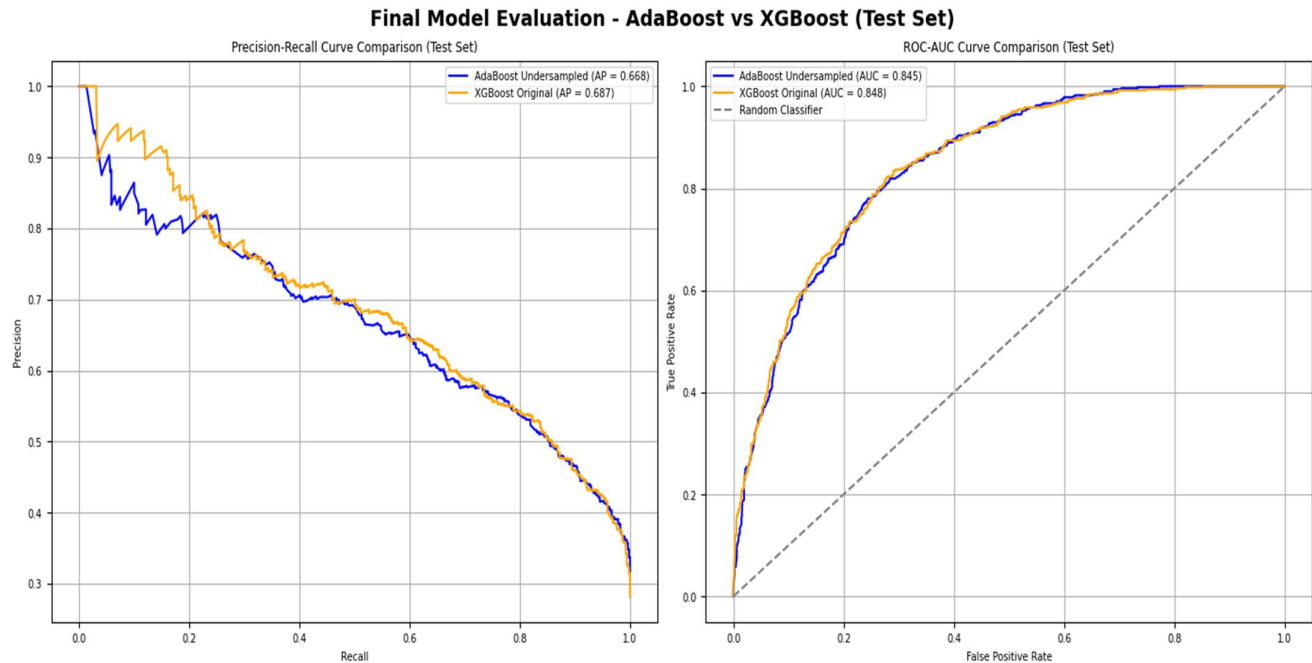


Fig G.4 – ROC & PR Curves

- Both models demonstrated strong classification capability with ROC-AUC values above 84%, indicating effective separation between churn and non-churn customers.

- XGBoost achieved slightly higher ROC-AUC and Average Precision scores, indicating better overall class discrimination and more balanced prediction performance.
- AdaBoost maintained comparatively stronger Recall performance, supporting its effectiveness in identifying potential churn customers.
- The Precision-Recall curves indicate that both models maintained stable Precision across different Recall levels, demonstrating robust classification capability on the test dataset.

Appendix G.5 Feature Importance Comparison

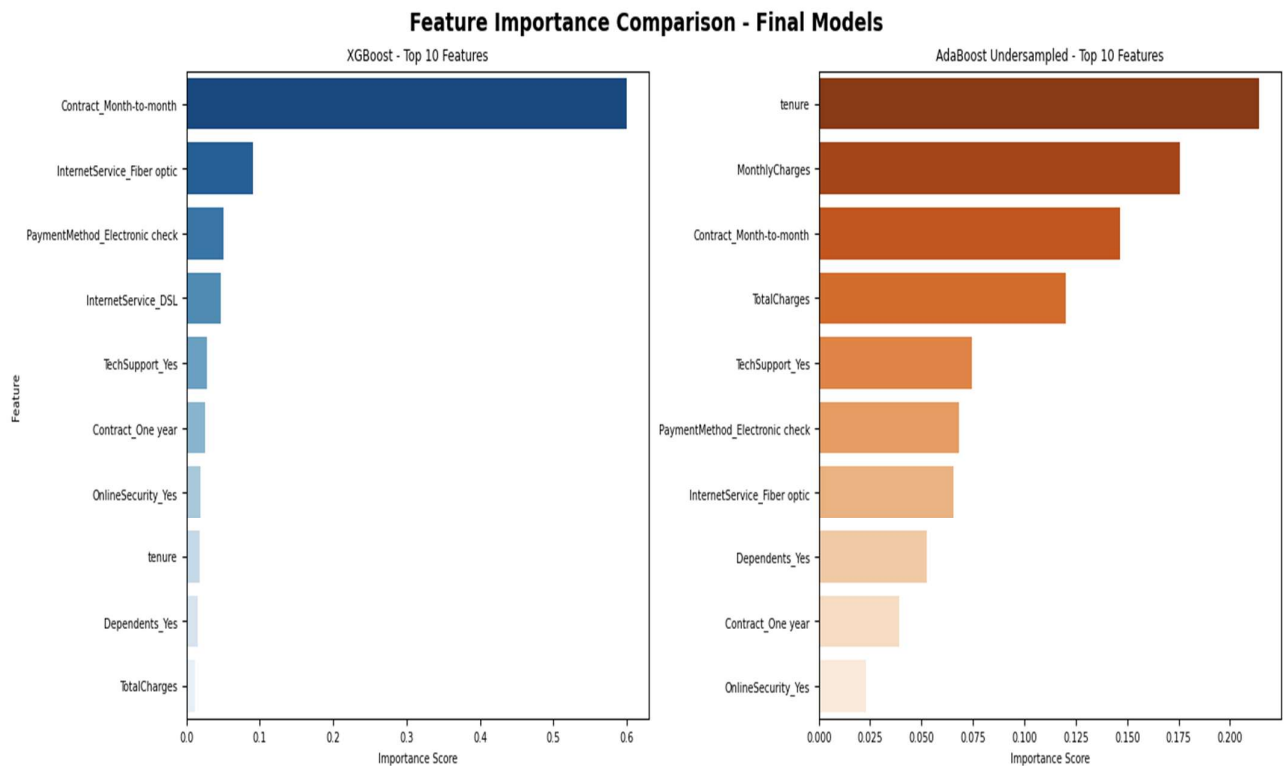


Fig G.5 – Feature Importance Comparison

- Contract type, tenure, monthly charges, and internet service type emerged as the most influential features across both final models.
- Both models identified month-to-month contracts and fiber optic internet service as strong indicators of customer churn.
- AdaBoost assigned higher importance to tenure and monthly charges, while XGBoost placed stronger emphasis on contract type.
- The feature importance analysis indicates that customer contract behavior, service type, and billing characteristics were key drivers of churn prediction.